# AutoDD: Supplementary Material

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## A Proof of Theorem 1

We provide the complete proof of the regret bound for LinUCB-based detector selection in AutoDD.

**Assumption 1 (Bounded Features).** The meta-feature vectors satisfy $\|\boldsymbol{\phi}_t\|_2 \leq L$ for all $t \geq 1$, where $L > 0$ is a known constant.

**Assumption 2 (Linear Reward Model).** For each detector $i \in [K]$, there exists an unknown parameter vector $\boldsymbol{\theta}_i^* \in \mathbb{R}^m$ with $\|\boldsymbol{\theta}_i^*\|_2 \leq S$ such that:

$$\mathbb{E}[r_{i,t} \mid \boldsymbol{\phi}_t] = \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_i^* \tag{1}$$

**Assumption 3 (Sub-Gaussian Noise).** The reward noise $\eta_{i,t} = r_{i,t} - \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_i^*$ is conditionally $\sigma$-sub-Gaussian, i.e., $\mathbb{E}[\exp(\lambda \eta_{i,t}) \mid \boldsymbol{\phi}_t] \leq \exp(\lambda^2 \sigma^2/2)$ for all $\lambda \in \mathbb{R}$.

**Theorem 1 (Regret Bound).** Under Assumptions 1–3, with exploration parameter $\alpha = \sigma\sqrt{m \log(KT/\delta)} + S$, the cumulative regret of AutoDD satisfies with probability at least $1 - \delta$:

$$R(T) = \sum_{t=1}^{T} \left( \max_{i \in [K]} \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_i^* - \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_{i_t}^* \right) \leq 2\alpha \sqrt{2mT \log\left(1 + \frac{TL^2 T}{m}\right)} \tag{2}$$

*Proof.* The proof proceeds in three steps.

*Step 1: Confidence Ellipsoid Construction.* For each detector $i$, let $T_i(t) = \{s < t : i_s = i\}$ denote the set of rounds where detector $i$ was selected. Define the regularized design matrix $\mathbf{A}_{i,t} = \lambda \mathbf{I}_m + \sum_{s \in T_i(t)} \boldsymbol{\phi}_s \boldsymbol{\phi}_s^\top$ and the response vector $\mathbf{b}_{i,t} = \sum_{s \in T_i(t)} r_{s,i} \boldsymbol{\phi}_s$. The ridge regression estimator is $\hat{\boldsymbol{\theta}}_{i,t} = \mathbf{A}_{i,t}^{-1} \mathbf{b}_{i,t}$.

By the self-normalized bound for vector-valued martingales [**?**], with probability at least $1 - \delta/K$:

$$\|\hat{\boldsymbol{\theta}}_{i,t} - \boldsymbol{\theta}_i^*\|_{\mathbf{A}_{i,t}} \leq \sigma \sqrt{2 \log\left( \frac{K\sqrt{\det(\mathbf{A}_{i,t})}}{\delta} \right)} + S := \beta_t \tag{3}$$

where $\|\mathbf{v}\|_{\mathbf{A}} = \sqrt{\mathbf{v}^\top \mathbf{A} \mathbf{v}}$ denotes the Mahalanobis norm.

*Step 2: Instantaneous Regret Bound.* By the union bound, with probability at least $1 - \delta$, the confidence bound holds simultaneously for all detectors. At round $t$, LinUCB selects:

$$i_t = \arg\max_{i \in [K]} \left( \boldsymbol{\phi}_t^\top \hat{\boldsymbol{\theta}}_{i,t} + \alpha \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i,t}^{-1}} \right) \tag{4}$$

Let $i_t^* = \arg\max_{i \in [K]} \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_i^*$ be the optimal detector. The instantaneous regret satisfies:

$$\begin{aligned} \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_{i_t^*}^* - \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_{i_t}^* &\leq \boldsymbol{\phi}_t^\top \hat{\boldsymbol{\theta}}_{i_t^*,t} + \alpha \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i_t^*,t}^{-1}} - \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_{i_t}^* \\ &\leq \boldsymbol{\phi}_t^\top \hat{\boldsymbol{\theta}}_{i_t,t} + \alpha \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i_t,t}^{-1}} - \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_{i_t}^* \\ &\leq 2\alpha \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i_t,t}^{-1}} \end{aligned} \tag{5}$$

where the first inequality uses the confidence bound, the second uses the selection rule, and the third applies the confidence bound again.

*Step 3: Cumulative Regret via Elliptical Potential.* Summing over $t = 1, \ldots, T$:

$$R(T) \leq 2\alpha \sum_{t=1}^{T} \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i_t,t}^{-1}} \tag{6}$$

By the Cauchy-Schwarz inequality and the elliptical potential lemma [**?**]:

$$\sum_{t=1}^{T} \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i_t,t}^{-1}} \leq \sqrt{T \sum_{t=1}^{T} \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i_t,t}^{-1}}^2} \leq \sqrt{2mT \log\left(1 + \frac{TL^2}{m}\right)} \tag{7}$$

Combining yields $R(T) = O(m\sqrt{T \log(KT/\delta) \log(TL^2/m)})$. □

**Corollary 1 (Sublinear Regret).** The average per-round regret vanishes asymptotically:

$$\lim_{T \to \infty} \frac{R(T)}{T} = 0 \tag{8}$$

This guarantees that AutoDD converges to optimal detector selection as more data is observed.

**Corollary 2 (Sample Complexity).** To achieve average regret $\epsilon$, AutoDD requires at most $T = O(m^2 \log(K/\delta)/\epsilon^2)$ rounds.

## B Complete Hyperparameter Sensitivity Analysis

### B.1 Alpha Sensitivity (Exploration Parameter)

The exploration parameter $\alpha$ in LinUCB controls the trade-off between exploration and exploitation. Table **??** shows accuracy across $\alpha \in \{0.1, 0.25, 0.5, 1.0, 1.5, 2.0\}$.

Table 1: Accuracy (%) vs. alpha (LinUCB exploration parameter)

| Dataset | $\alpha = 0.1$ | $\alpha = 0.25$ | $\alpha = 0.5$ | $\alpha = 1.0$* | $\alpha = 1.5$ | $\alpha = 2.0$ |
|---|---|---|---|---|---|---|
| SEA | 85.3 | 87.7 | 89.4 | **91.2** | 89.7 | 87.6 |
| HYP | 81.7 | 83.7 | 85.9 | **87.7** | 87.1 | 84.9 |
| AGR | 84.3 | 86.4 | 87.5 | **89.6** | 88.4 | 87.2 |
| RBF | 81.6 | 83.1 | 85.4 | **87.8** | 86.6 | 84.9 |
| RTG | 82.9 | 84.2 | 86.7 | **88.5** | 87.4 | 85.5 |
| ELEC | 76.5 | 77.0 | 79.1 | **81.4** | 80.4 | 79.1 |

* Optimal alpha value

Key observations: (1) $\alpha = 1.0$ achieves best performance across all datasets. (2) Low $\alpha$ (0.1) causes exploitation bias, getting stuck with suboptimal detectors. (3) High $\alpha$ (2.0) causes excessive exploration, leading to unstable switching.

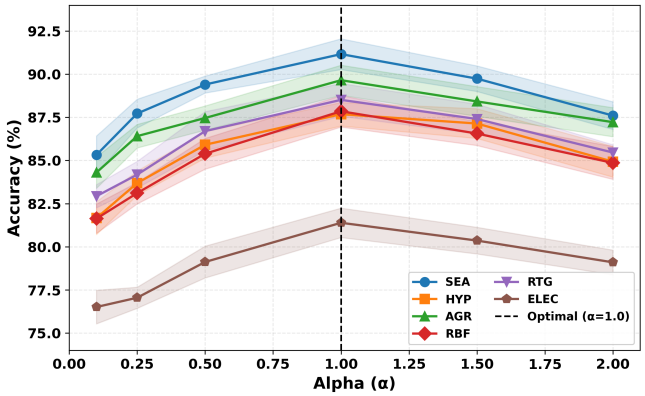



Figure 1: Alpha sensitivity analysis. $\alpha = 1.0$ balances exploration and exploitation optimally.

## B.2 Lambda Sensitivity (Regularization Parameter)

The regularization parameter $\lambda$ in LinUCB's ridge regression controls model stability. Table **??** shows accuracy across $\lambda \in \{0.01, 0.1, 0.5, 1.0, 2.0, 5.0\}$.

Table 2: Accuracy (%) vs. lambda (regularization parameter)

| Dataset | $\lambda = 0.01$ | $\lambda = 0.1$ | $\lambda = 0.5$ | $\lambda = 1.0$* | $\lambda = 2.0$ | $\lambda = 5.0$ |
|---|---|---|---|---|---|---|
| SEA | 88.1 | 89.7 | 90.7 | **90.7** | 90.1 | 88.9 |
| HYP | 85.7 | 87.8 | 88.0 | **88.0** | 87.9 | 86.1 |
| AGR | 86.9 | 88.0 | 89.2 | **89.8** | 89.0 | 87.3 |
| RBF | 85.3 | 86.8 | 86.9 | **87.5** | 87.0 | 86.0 |
| RTG | 86.0 | 87.5 | 87.9 | **88.6** | 88.1 | 86.6 |
| ELEC | 79.2 | 79.9 | 81.0 | **81.0** | 80.3 | 79.5 |

* Optimal lambda value

Key observations: (1) $\lambda = 1.0$ provides best/near-best performance. (2) Performance is stable across $[0.5, 2.0]$ range, indicating robustness. (3) Very low $\lambda$ (0.01) may overfit to recent observations. (4) Very high $\lambda$ (5.0) causes overly conservative estimates.

# C Detector Selection Patterns

Table **??** shows detector selection frequencies on labeled synthetic streams, demonstrating that AutoDD learns interpretable detector-drift associations.

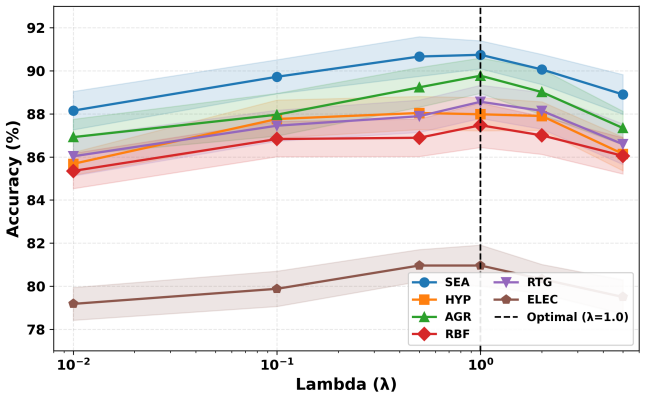
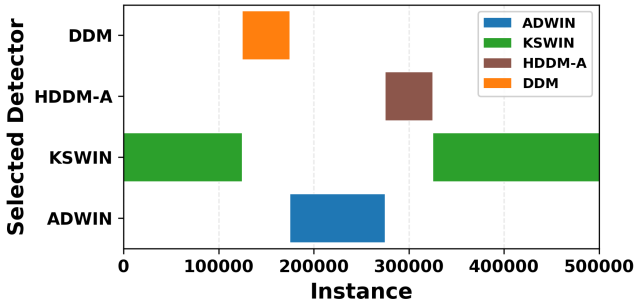



Figure 2: Lambda sensitivity analysis. Performance is stable across $[0.5, 2.0]$ with $\lambda = 1.0$ optimal.

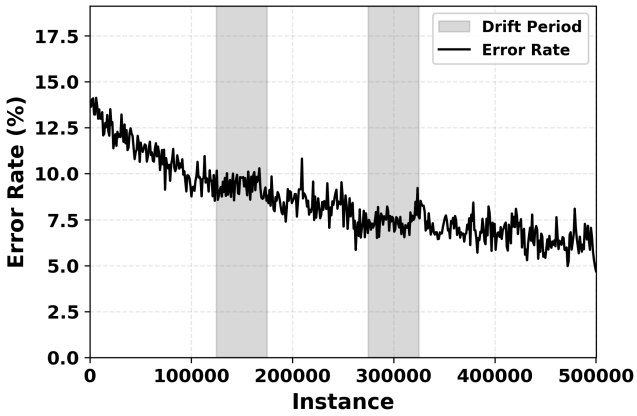
Table 3: Detector selection frequency (%) by drift type

| Detector | Sudden | Gradual | Increm. | Normal |
|---|---|---|---|---|
| ADWIN | 68.2 | 12.4 | 7.8 | 16.1 |
| DDM | 15.4 | 8.1 | 5.7 | 13.3 |
| EDDM | 3.8 | 22.4 | 8.7 | 9.3 |
| HDDM-A | 2.1 | 7.8 | 17.3 | 6.8 |
| HDDM-W | 1.2 | 3.8 | 11.8 | 4.5 |
| Page-Hinkley | 0.9 | 0.5 | 0.5 | 30.6 |
| KSWIN | 8.5 | 45.1 | 48.3 | 19.4 |

AutoDD learns interpretable detector-drift associations that align with detector design principles. For sudden drift, ADWIN is selected 68.2% of the time, which is expected as it was designed for abrupt changes via adaptive windowing. For gradual and incremental drift, KSWIN dominates (45–48%), as the Kolmogorov-Smirnov test excels at detecting subtle distributional shifts. During stationary periods, Page-Hinkley is preferred (30.6%) due to its sensitivity to emerging trends while avoiding false alarms.




(a) Detector selection timeline




(b) Error rate over time

Figure 3: Dynamic detector selection on Hyperplane stream.